# LAKSHMI K

## PRODUCTION ENGINEER

Thrissur, Kerala • +91 7736245162 • lakshmikdev08@gmail.com
https://www.linkedin.com/in/lakshmi-k-756931229/

## SUMMARY

M.Tech student in Manufacturing Technology at IIST with a B.Tech in Production Engineering and experience in advanced manufacturing, CAD, molecular dynamics simulation, and engineering analysis. Proficient in SolidWorks, AutoCAD, LAMMPS, OVITO, MATLAB, and manufacturing processes. Seeking opportunities in production, manufacturing, mechanical design, process engineering, or aerospace manufacturing.

## EDUCATION

**M.TECH IN MANUFACTURING TECHNOLOGY** | IIST

Relevant Coursework: Plasticity & Metal Forming • Additive Manufacturing • Composite Manufacturing • Welding Technology • Molecular Dynamics • Quality Engineering • CAM & Manufacturing Planning
Admission Year: 2025 | Current CGPA: 7.47

**Bachelor of Technology in Production Engineering**

Government Engineering College, Thrissur | CGPA: 7.07
Relevant Coursework: CAD/CAM • Production Processes • Metal Forming • Metrology • Industrial Robotics • Additive Manufacturing • Operations Management

## RESEARCH EXPERIENCE

**M.Tech Research Internship – Molecular Dynamics Simulation** **2026**
Department of Aerospace Engineering, IIST | Thiruvananthapuram

- Reconstructed and validated an all-atom gas detonation forming model in LAMMPS.
- Studied aluminium deformation under different high-speed punch velocities.
- Used OVITO to analyse displacement, defects, fracture and material fragmentation.

## WORK EXPERIENCE

**Mechanical Engineering Intern** **March - July 2025**
Nava Design & Innovation Pvt. Ltd.

- Developed detailed 3D models and engineering drawings using SolidWorks, ensuring high accuracy for prototyping.
- Gained hands-on experience in production workflows and coordinated with vendors for component procurement.
- Assisted in the testing and documentation of new products, enhancing analytical skills through data interpretation.

**Internship Trainee** **May 2023**
HMT Machine Tools Limited | Ernakulam, India

- Gained foundational knowledge of manufacturing processes, from metal casting (foundry) to precision machining.
- Analysed assembly drawings and the integration of mechanical components, enhancing proficiency in reading and interpreting engineering drawings.
- Observed Quality Control methods applied to high-precision CNC turning centres.

---

## PROJECTS

### Semi-Automation of Koragrass Looming Machine

- Designed a cost-effective automation mechanism for traditional looms.
- Integrated mechanical linkages to improve production speed.

### FORMULA BHARAT 2023

Collaborated on the design and hands-on fabrication of suspension components, ensuring compliance with strict Formula Bharat technical regulations.

### IoT-Based Drip Irrigation System

Developed a smart irrigation prototype using sensors and controllers.

---

## CORE COMPETENCIES

- Software & Simulation: LAMMPS, OVITO, Atomsk, MATLAB
- CAD & Design: SolidWorks, AutoCAD, Creo, Autodesk Fusion
- Manufacturing: CNC Programming, Machining, Metrology, Foundry Processes, Computer-Aided Manufacturing
- Engineering: Production Planning, Manufacturing Planning & Control, Procurement, Process Improvement
- Emerging Technologies: Foundations of Machine Learning
- Soft Skills: Communication, Team Coordination, Time Management, Multitasking
- Languages: English, Malayalam

---

## INDUSTRIAL VISITS

- Vikram Sarabhai Space Centre (VSSC)
- Liquid Propulsion Systems Centre (LPSC) – Thiruvananthapuram
- ISRO Inertial Systems Unit (IISU)
- Steel and Industrial Forgings Limited (SIFL) – Aerospace & Defence Forgings
- Instrumentation Ltd, Palakkad (ILP)

---

## CERTIFICATIONS

NPTEL: Product Design and Development IIT Roorkee (Score: 88%)